

Assignment - 2

Section A

Solⁿ 1) CPU registers are small, high-speed storage locations inside the processors

Importance:

- Store intermediate data & instructions
- faster than cache & memory
- improve execution speed of programs

Solⁿ 2) Program Counter (PC) is a register that holds the address of the next instruction

Function:

- Points to the next instruction to be executed
- Automatically update after each instruction
- Controls instruction sequence

Solⁿ 3) Instruction format defines the structure of an instruction in binary form

- Key fields:
- opcode (operation)
 - operand
 - Addressing mode

Solⁿ 5) Addressing mode specify how to locate operands

Purpose:

- Provide flexibility in accessing data
- Reduce instruction count
- Improve program efficiency

Solⁿ 6) ISA is the interface between hardware & software defining supported instructions

Includes:

- Instruction formats
- Data types
- Registers & addressing modes

Solⁿ 7) Types of instructions in a processor are:

- Arithmetic Arithmetic (ADD, SUB)
- Logical (AND, OR)
- Data Transfer (LOAD, STORE)
- Control (JUMP, BRANCH)

Solⁿ 8) Explanation:

- Opcode is interpreted
- Required operation is identified
- Control signals are generated
- Operands are prepared for execution

Solⁿ 9) Basic instruction cycle:

- fetch instruction from memory
- ~~Decode instruction~~ instruction
- ~~execute~~ execute operation
- store result

Solⁿ 10) Characteristics of RISC processors are:

- simple & fixed-length instructions
- few addressing modes
- faster execution (pipeline-friendly)
- load/store architecture

Solⁿ 11) Characteristics of CISC processors are:

- complex & variable-length instructions
- many addressing modes
- fewer instructions per program
- slower but powerful instructions

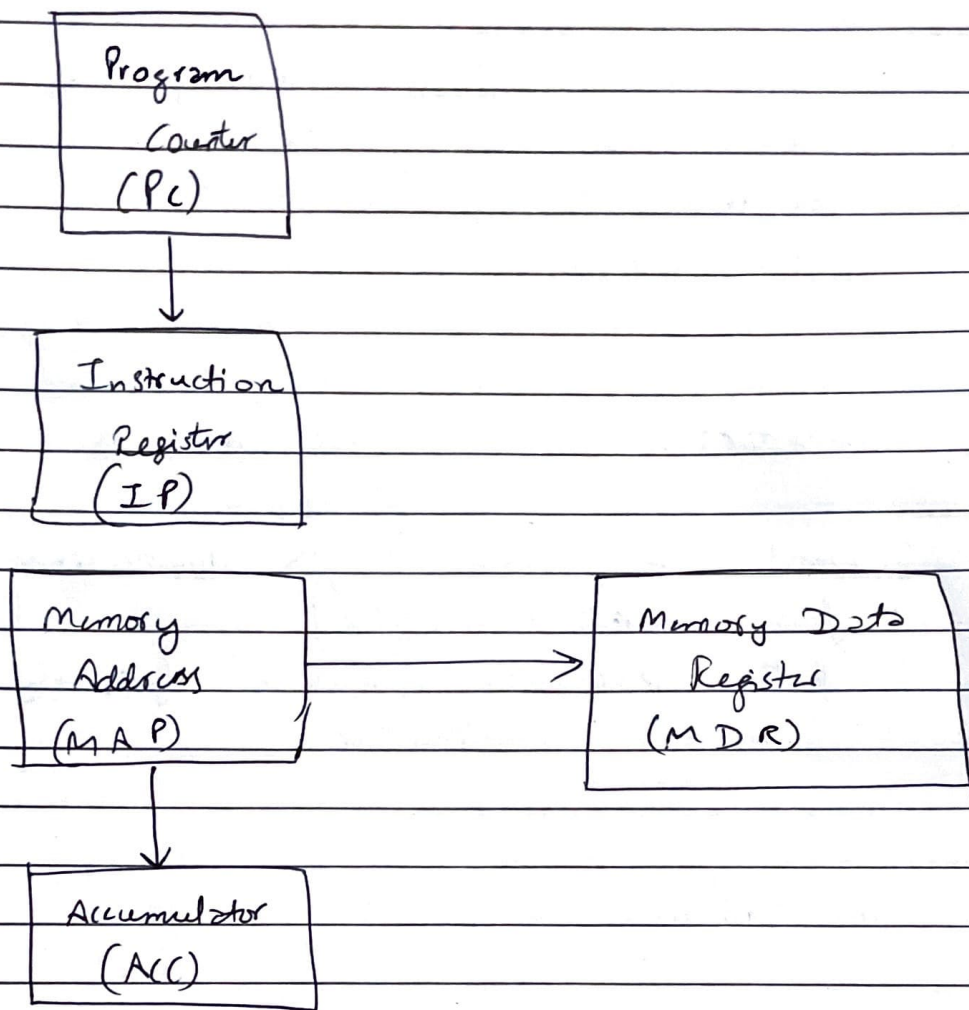
Solⁿ 12) RISC-V:

- open source ISA
- simple, fixed length instructions
- easier to design & extend

x86:

- ~~propriety~~ proprietary (Intel / AMD)
- complex variable length instructions

Section - B

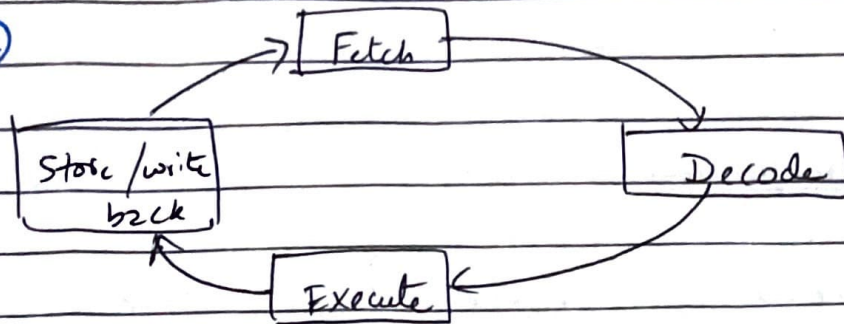
(67th 1)

Explanation:

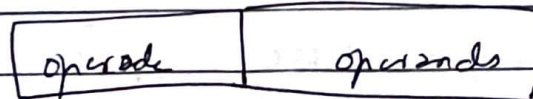
- PC holds next instruction address; IR stores current instruction
- MAR & MDR handle memory address; Accumulator stores immediate results

(P.T.O.)



Solⁿ 2)

- CPU fetches instructions, decode it, execute it & write back

Solⁿ 3)

- Explanation:
- opcode specifies the operations
 - operands specify data or memory location

Solⁿ 4)

Feature	RISC	CISC
Instruction	Simple	Complex
Length	fixed	variable
Speed	fast	slower
Memory	more usage	less usage

- Explanation:
- RISC focuses on simplicity & speed
 - CISC focuses on power & complexity

Section 7

Solⁿ 1) Impact on performance are:

- Memory access is slower than register access
- Increases CPI due to memory latency
- Overall execution time increases → performance decrease.

Solⁿ 2) The should choose RISC architecture because:

- Simple & fixed-length instructions
- Easier to pipeline efficiently
- Requires simpler hardware design

Solⁿ 3) CPU executes the instructions as:

Step 1: Fetch:

- Instruction is fetched from memory into IR

↗

Step 2: Decode:

- Opcode is decoded & registers are identified

Step 3: Execute:

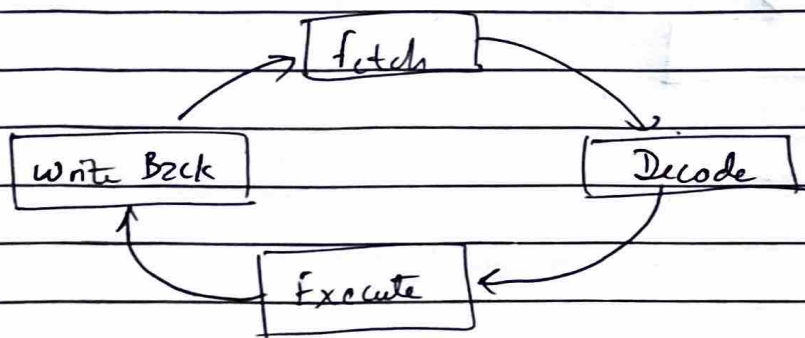
- Data is read from registers
- ALU performs operation
- Result is stored back in register

solⁿ 4) X86 is widely used despite complexity because:

- Backward compatibility with ~~other~~ older software
- Large ecosystem (OS, apps, ~~ind~~ industry support)
- Modern CPU internally optimized
- Despite complexity, compatibility & ecosystem keeps the X86 dominant.

Section - D

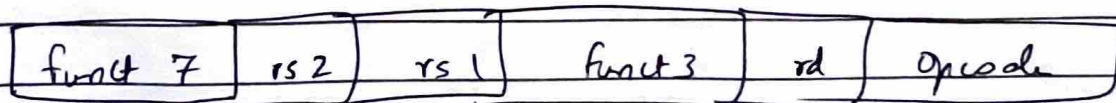
solⁿ 1)



Explanation of fetch stage:

- Program counter (PC) holds address of next instruction.
- Address sent to memory via MAR
- Instruction fetched into MPR & loaded into IR
- Program counter is incremented to next instruction

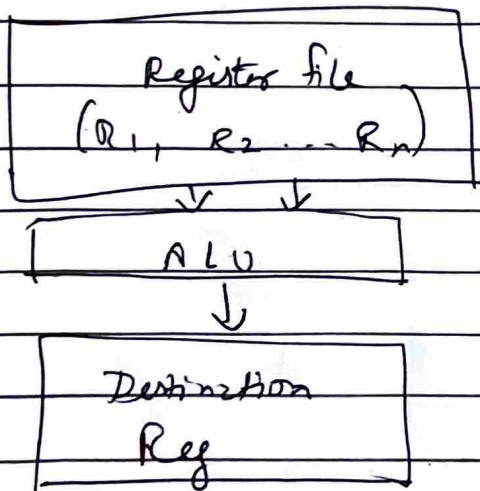
solⁿ 2)



(P. 7)

Explanation:

- opcode - specifies operation type
- rd → destination register
- rs1, rs2 → source registers
- funct3, funct7 - Define exact operation
- fixed length (32 bit) format simplifies design

Sol³ 3)

Explanation:

- Data is read from source registers
- Sent to ALU for operation
- Result is written back to destination register
- Enable fast communication without memory access